



Waipapa
Taumata Rau
**University
of Auckland**

Social Cost Benefit Analysis – Valuing what matters

Lecture 5

➤➤➤ 30 July, 2026

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Learning Outcomes

1. Explain how social cost-benefit analysis supports evidence-based government decision making. Identify and value the economic and societal impacts of systems-level projects.
2. Apply social cost-benefit analysis to evaluate competing project options.
3. Interpret financial and non-financial evidence to justify investment decisions.
4. Evaluate the limitations of monetising societal outcomes and the importance of broader wellbeing considerations.

Outline

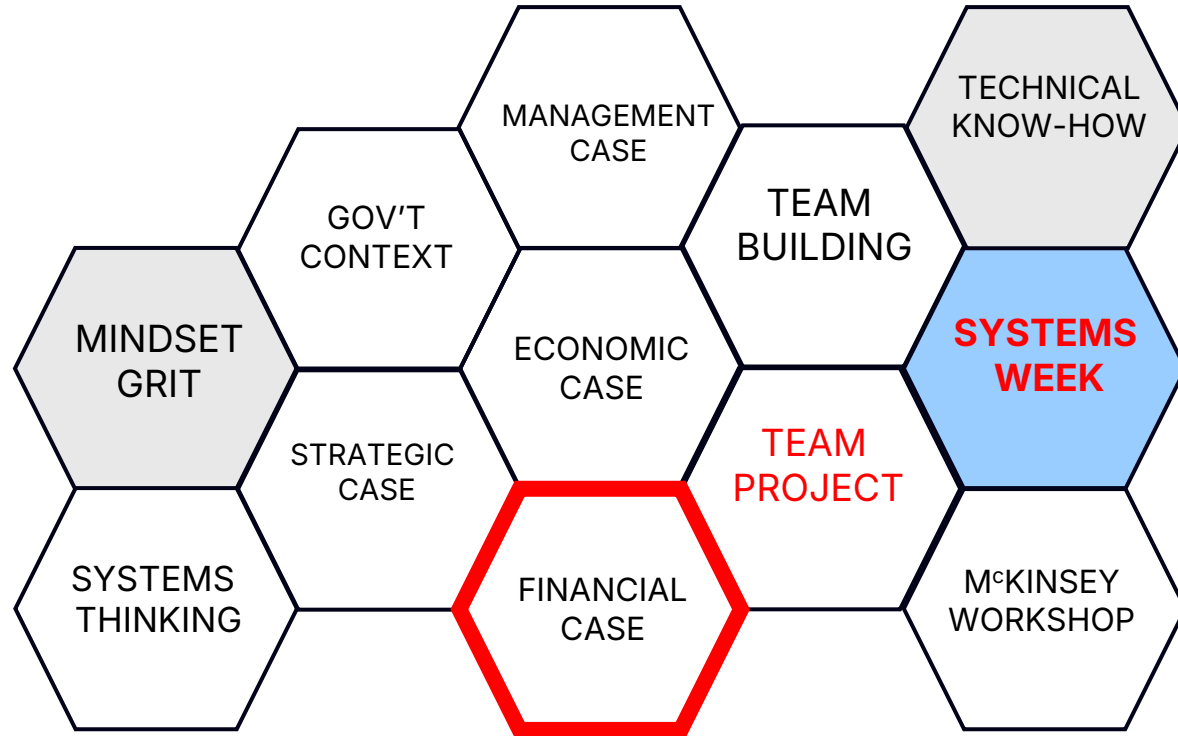
1. Recap from ENGGEN 303
2. Why Social Cost-Benefit Analysis?
3. Identifying and Valuing Impacts
4. Monetising What Matters
5. Making Better Investment Decisions
6. Limits of Cost-Benefit Analysis
7. Summary and Questions

See [canvas page](#) for more business cases



VS





ENGGEN 303

ENGGEN 403

Problem space

Strategic case

- Project background/Introduction
- Problem statements
- Users/stakeholders
- Requirements and CSFs

Economic Case

- Options
- Long list options assessment
 - DFV
 - Problem statements and CSFs
- Shortlisted options

Solution space

Financial Case

- Economic and social impact assessment of at least 2 shortlisted options
- Preferred way forward

Implementation plan

Management Case

- Project scope
- Project plan

Cost Benefit Analysis Recap



Benefits and Costs

In ENGGEN 303 we covered **discounted cash flow** and **cost benefit analysis** for a single project/product.

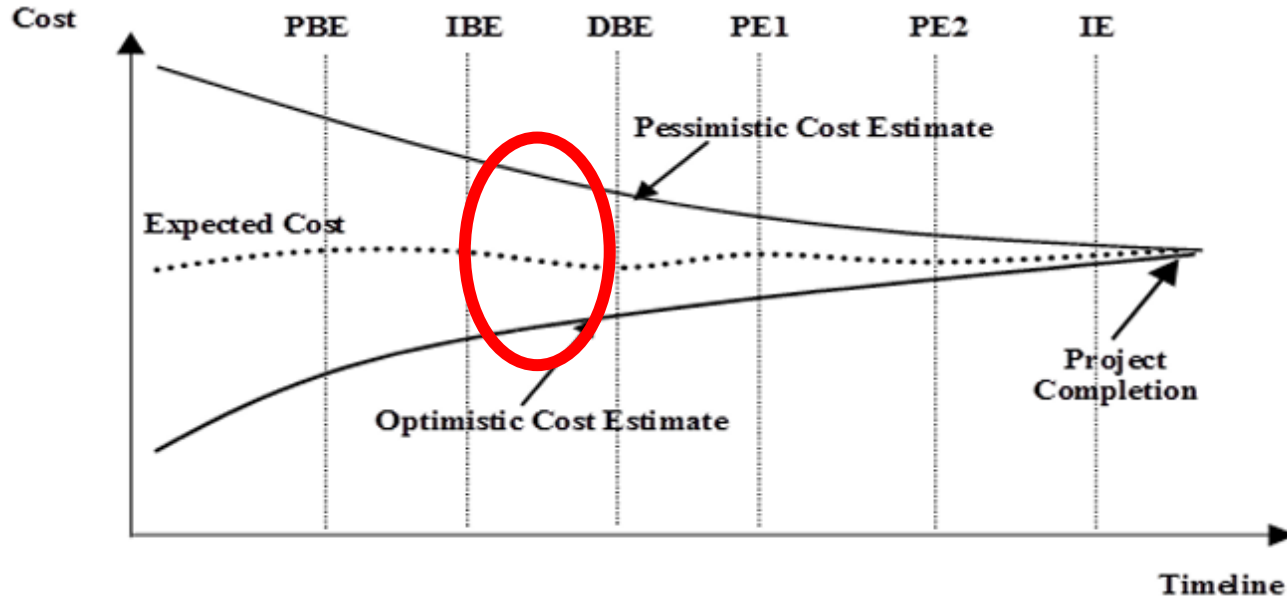
- Income (revenue)
- Expenses (costs)



System-level projects use the **social cost benefit analysis** as these projects also

- Generate social/economic benefits (e.g. less congestion, reduced hospital visits)
- Have social/economic costs (e.g. increased emissions, ecology impact of a new dam)

Costs



- Programme business case estimate (PBE)
- Indicative business case (IBE)
- Detailed business case estimate (DBE)
- Pre-implementation estimate 1 (PE1)
- Pre-implementation estimate 2 (PE2)
- Implementation estimate (IE)

Costs

Transport Infrastructure

[NZTA Cost Estimation Manual](#)
[NZTA Monetised benefits and costs manual](#)

[NZTA Financial Case](#) (tools at the bottom of page)
[NZTA other useful tools](#)

[AT guide to cost estimation](#)

System project costs are high level, order of magnitude:

Capital costs:

- Land acquisition (if needed)
- Construction
- Equipment

Ongoing costs:

- Labour
- Operations
- Utilities

Figures rounded to 2 SF reflects uncertainty!

Will need to make assumptions - look at studies, reports, similar projects, and other resources

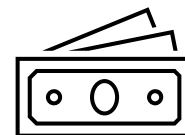
Package 1: Do Minimum		Quantifying Options 2, 5, 6, 8
Quantified Effect (cost): Container return scheme		
Assumption(s):		
A	The cost of each machine is \$40,000 (Davies, Barton, 2022).	
B	1 machine is sufficient to service 20,000 people on average.	
C	It costs \$14,672 required to lease machines for companies to use per year.	
Calculations:		
Total number of machines = 5.3 million people / 20,000 = 267 machines		
Total Upfront cost: \$40,000 × 267 = \$10.8 million		
Total Ongoing cost: \$14,762 × 267 = \$3.94 million		
The total upfront cost of the of the container return scheme will be \$11 million.		
The total upfront cost will be covered during the 1 st year. Ongoing cost will be covered from 2 nd year onwards until the 5 th year.		
The total cost of the container return scheme will be \$15 million.		
Quantified Effect (cost): Advertising campaign		
Assumption(s):		
A	Campaign reaches similar scale to the <i>Keep it Real Online</i> advertising campaign. (Keep It Real Online, 2020)	
Calculations:		

Time value of money and discount rates



Recap 303

The time value of money states that money is worth more in the present than the same amount in the future because you can invest it in the interim.



Economic Metrics

Net Present Value:

$$NPV = \left(\sum_{n=1}^n \frac{\text{Undiscounted Benefits} - \text{Undiscounted Costs}}{(1+r)^n} \right)$$

r = discount rate
 n = time period

$NPV > 0$ – Accept project

$NPV < 0$ – Reject project

Benefit-Cost Ratio:

$$BCR = \frac{PV_{benefits}}{PV_{costs}}$$

Cannot use one of these in isolation to determine Go/No-go

$BCR > 1$: Accept project

$BCR < 1$: Reject project

Internal Rate of Return:

$$0 = \left(\sum_{n=1}^n \frac{\text{Undiscounted Benefits} - \text{Undiscounted Costs}}{(1+r)^n} \right)$$

$IRR > 0$ – Accept project

$IRR < 0$ – Reject project

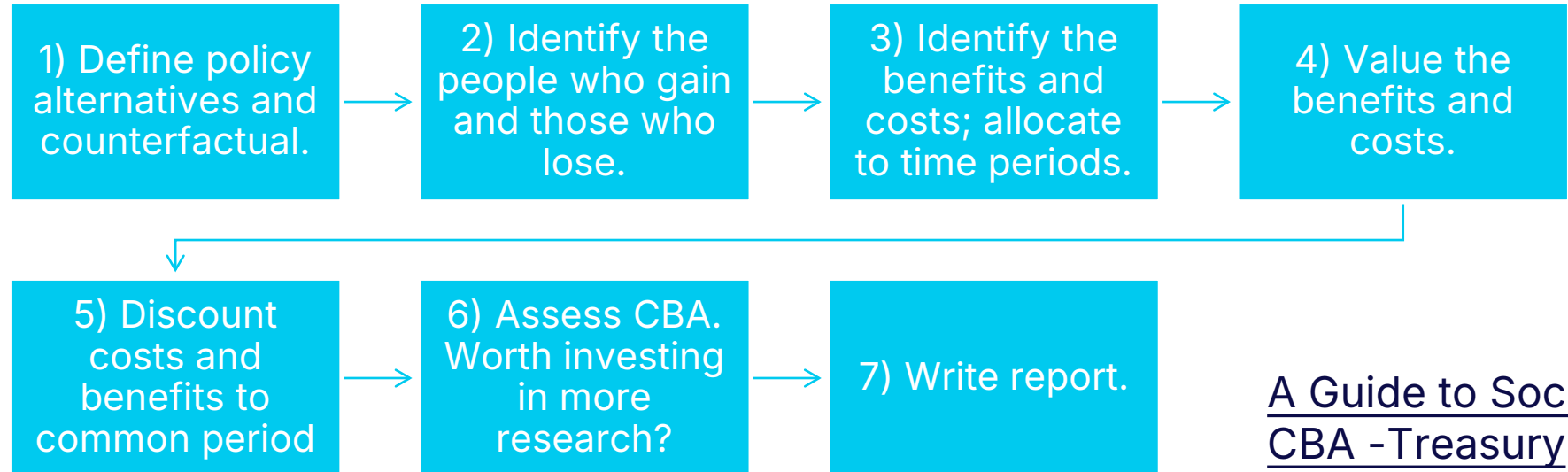
Social Cost Benefit Analysis



Social Cost Benefit Analysis

Once options assessment is complete and you have a shortlist you must now assess the **Financial Case** for these options. One way to do this is the **social cost benefit analysis**

At the systems (national) level the steps for the CBA are:



A Guide to Social
CBA -Treasury

Step 1. Identifying the Counterfactual

Example: Bridge over river

Suppose that the bridge costs \$20 million, and that it will save travellers \$25 million worth of travel time and vehicle operating costs, in present value terms. The bridge would appear to have benefits that exceed the costs. The net present value (NPV) of the bridge is \$5 million.

But suppose that in the absence of a bridge being built, there is every expectation that a private ferry operator will start business. The cost is \$10 million in present value terms, and the social benefits are \$20 million in present value terms. The ferry operation has an NPV of \$10 million.

Compared with the ferry operation, a bridge would cost \$10 million more, and would produce \$5 million more benefits. Against this counterfactual, the bridge has an NPV of $-\$5$ million.

Against the “no bridge, no ferry” counterfactual, the bridge would seem worthwhile. But against the “ferry” counterfactual, the bridge is not.

Equivalently, the ferry could be presented to decision-makers as an alternative to the bridge. This would still show the ferry to be the better option, despite the fact that the bridge has greater total benefits.

Options against the counterfactual

SYSTEMIC INTERVENTIONS FOR PLASTIC POLLUTION IN AOTEAROA

ENGGEN 403 SYSTEMS THINKING

OCTOBER 3 2025



Need to define your baseline counterfactual. AKA Do nothing.

- Do nothing more? (business as usual)
- Do less? (roll back initiatives)

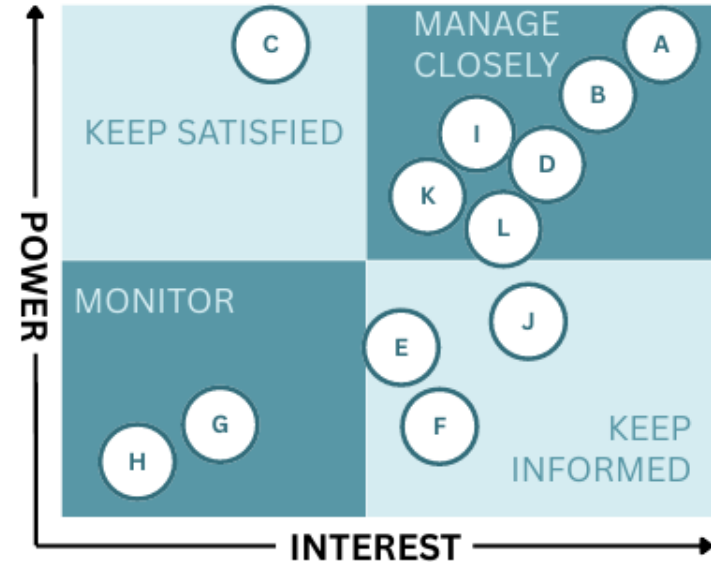
3.4.1 Do Nothing Case

The Do Nothing preserves current-day conditions to help illustrate the benefit of additions. It establishes the reference case for CBAX comparisons and helps quantify the cost of inaction, including lost revenue and increasing landfill pressures. While this package has no operating costs, over time it results in higher council liabilities, greater environmental degradation, decrease in societal wellbeing and reduced revenue potential. Including this package is essential to demonstrate the incremental value of subsequent options and to ground the overall cost-benefit analysis.

Step 2. Identify who gains and who loses

Stakeholder analysis. Who are the people these national level projects will impact?

Refer to Lecture 7: Stakeholders at the systems level



Steps 3 & 4. Identify and value benefits and costs

3 Description of Benefits

Monetary: Benefits that are measurable in monetary terms: financial or cash releasing, such as avoided costs or efficiency savings

Non-monetary: Benefits that are not measurable in monetary terms. These can either be quantifiable, such as reduced customer complaints, or qualitative, such as improved health outcomes. Qualitative benefits may be observed but not easily measured. Narrative and level of change identified.

Identify benefits and costs

Identify impacts on people

e.g. reduced congestion increases transport efficiency, reduces travel times which also reduce driver frustration.



Quantify benefits and costs

Quantify by how much congestion therefore how much time saved. Carbon emissions avoided, vehicle operating cost saved



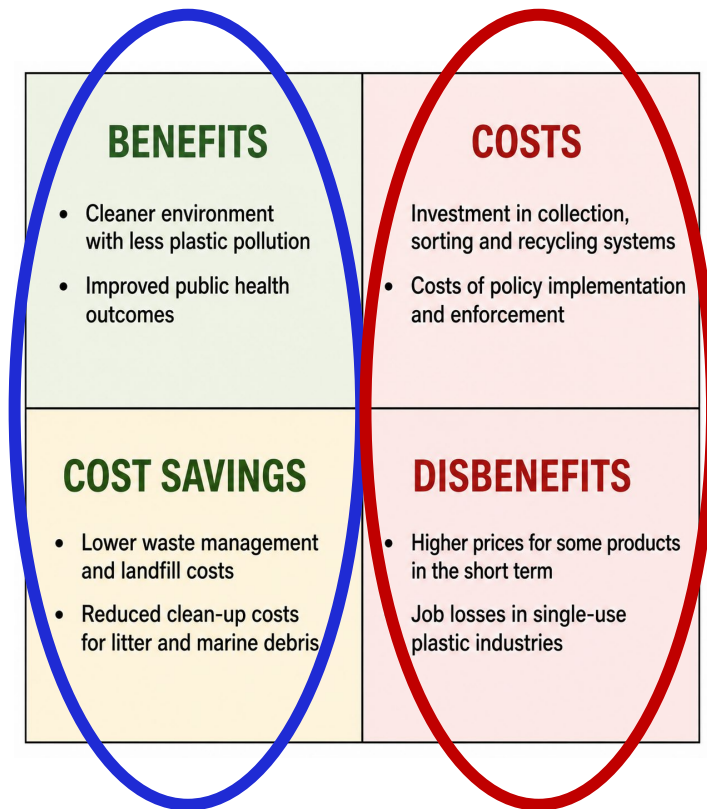
Monetise benefits and costs if possible

Monetise what has been quantified through metrics like the Value of Travel Time Savings (VTTs). Emissions trading scheme, CBAX

Benefits, costs, disbenefits, and cost savings

Solutions (options) can generate four types of outcomes:

- **Benefits** – positive outcomes for society. Including revenues
- **Disbenefits** – negative outcomes for society.
- **Costs** – money the government must spend to deliver and operate the option.
- **Cost savings** – money the government avoids spending compared with the *do nothing* case.



The Value of a Human Life

Price of life: Govt to value safer and faster journeys nearly three times more

- 2018 -report recommended removing car parks leading into roundabout, was not implemented.
- 2022 -Driver opens door, cyclist thrown in front of oncoming traffic. Dies at the scene.
- As engineers you may make decisions around changes to infrastructure that impact PEOPLE.
- Need to weigh safety and damages against things like convenience.



Image - Royal oak roundabout

Read the full story: <https://newsroom.co.nz/2023/05/01/govt-to-pay-three-times-the-price-for-faster-safer-journeys/>



The VOSL is **not the value of an identifiable person's life**. It is an economic measure of how much society is willing to pay for small reductions in mortality risk across a population. It is commonly used in social cost-benefit analyses of transport, environmental, and public health projects.

The Value of a Human Life

How do we value a human life?

- NZTA – value of a statistical human life
- 700-person survey **1991 = \$2M NZD**
- Doubled in **1998 = \$4M NZD**
- Before COVID Inflation pushed it to **\$4.53M NZD**
- 8000-person survey **2023= \$12.5M NZD**

Weighed against the value of convenience and time savings.

- Saving transport time \$7.80/hr (pre-2023) to **\$19.53/h** (post-2023)
- Increases to **\$36.18/hr** to not be stuck in congestion.

The amount of hours save from time savings multiplied by \$36.18

Willingness to Pay/Accept (WTP/WTa)

The willingness of someone to pay to avoid a bad outcome or the amount they would accept to give something up.

- Surveys e.g. contingent valuation, NZ General social survey, Overall life satisfaction
- Wellbeing valuation method
- Quality adjusted life years
- Subjective wellbeing valuation
- Market studies
- Wellbeing adjusted life year (WELLBY)
- Hedonic Pricing

[UK Greenbook guidance on wellbeing](#)
[Australian Social Value Bank](#)

Auckland

NZ General Social Survey: 2014



Highly rated overall life satisfaction

82.7% Auckland

82.6% New Zealand



Key points for Auckland region



40.5 percent said they didn't need home repairs or maintenance.



Highest percentage with excellent health.



Nearly two-thirds hadn't felt lonely in the past four weeks.



Highest percentage with not enough money.

Read more from Treasury on WTP and valuation methods [here](#)

Identifying Impacts



Diagram 1. CBA analysis of impacts - only a subset of the impacts are monetised

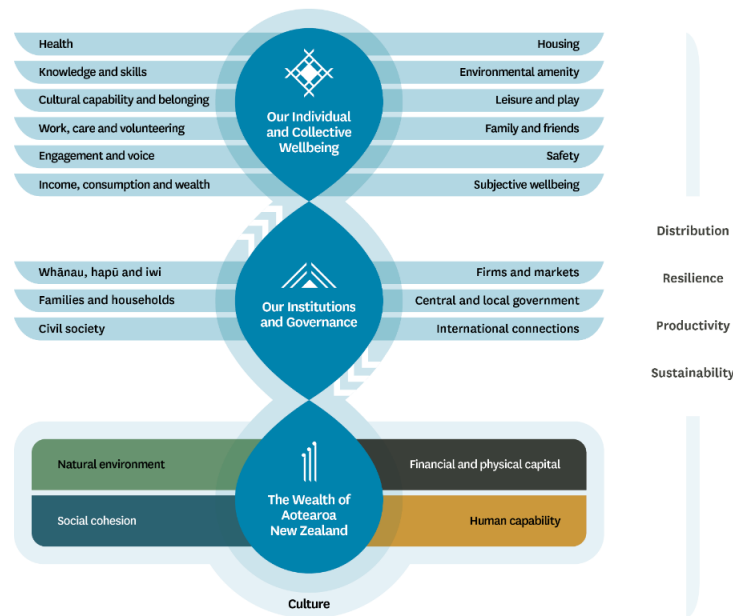
CBA Steps	1. Identify impacts - Comprehensive - Unquantified	2. Quantify impacts - To extent/where possible - Quantified but un-monetised	3. Monetise impacts - Selective - Quantified and monetised
Analysis	• Problem/opportunity and intervention logic • Counterfactual and options • Broad range of impacts: positive/negative, people, wellbeing domains, sectors, government/non-government, total economic value • Qualitative judgements	• Assumptions relative to counterfactual: Who are affected? In what way? To what extent? When? For how long? • Transparent quantification • Evidence base • Quantitative, e.g. could be used in cost-effectiveness (one main objective) or cost-utility analysis	• Monetise key impacts • Sensitivity analysis • Discount future impacts • Calculate overall net present value (NVP) across all monetised impacts and benefit/cost ratio (BCR) • If evidence is weak, use reverse CBA analysis
In practice	• Minimum analysis to give a sense of whether an option is doing some good • Check whether too narrow • Other perspectives?	• Difficult and time consuming • Involves research / modelling • Start with what you know • Stronger evidence base gives more confidence	• Model present value (PV) • Focus on PVs for key impacts, less on summary measures • CBAX tool includes database of monetary values
Measures	• Multi-dimensional wellbeing such as Living Standards wellbeing domains • Safety/Health: "Reduced road deaths"	• Quantified assumptions and evidence base • Marginal "number of prevented road deaths", types, settings etc.	• PV of impacts in dollar terms, using quantified analysis and impact values in modelling • Summary measures: NPV, BCR, and return-on-investment

Identifying Impacts

Need to identify impacts to do with society, such as the environment, health, education, quality of life etc.

Think:

What societal impacts will this option have?



Living Standards Framework



Monetising Impacts

CBAX spreadsheet

The CBAX spreadsheet tool is a spreadsheet model that contains a database of values for monetising impacts on the 12 domains of wellbeing.

- **Allows for consistent approach to cost benefit analysis through common assumptions and values.**
- **Long term and broad view of societal impacts, costs and benefits.**
- **Monetise and discount impacts where possible**
- **Transparency around assumptions and evidence base**

Using the CBAX Spreadsheet

Check the **Impacts database**, are impacts relevant to your project?

- Try to monetise the impacts (1-3 primary impacts) you are trying to improve (problem statement).
- CBAX your best bet, can try other tools.
 - Waka Kotahi Monetised Benefits and Costs Manual
 - Project emissions estimation tool (PEET)
- Otherwise, a bit of research into primary and secondary resources.

Full worked example, page 24 onwards, [CBAX guide](#)

Monetising Impacts

Impact database

Over 250 monetised impacts. Check the Impacts Database Tab.

[See it here](#)

Impacts Database

Take note of the row number (column A) for the relevant impact(s) values you want to enter in the CBAx Tool. See the CBAx Tool User Guidance for more information about entering values.

Row Number	Wellbeing Domain	Description	Value adjusted to 2024	Value	Unit	Government/Non-Government
129	Health	Cost to health system from fatal crash	-17,149	-16,200	Per incident	Government
130	Health	Cost to health system of serious crash	-20,642	-19,500	Per incident	Government
131	Health	Physical health gain from walking	10	10	Per pedestrian km	Non-Government
132	Health	Physical health gain from cycling (conventional)	5	5	Per pedestrian km	Non-Government
133	Health	Physical health gain from cycling (electric-assisted)	3	3	Per pedestrian km	Non-Government
141	Health	Dementia care	-240	-170	Per day	Government
142	Health	Hospice care	-972	-690	Per day	Government
143	Health	Inpatient hospital visit	-7,488	-6,530	Per visit	Government
144	Health	Outpatient hospital visit	-493	-350	Per visit	Government
145	Health	Intensive care unit	-8,318	-5,500	Per day	Government
146	Health	Emergency room	-521	-370	Per visit	Government
147	Health	Specialist visit (initial)	-493	-350	Per visit	Government
148	Health	Specialist (subsequent)	-352	-250	Per visit	Government
149	Health	Specialist (per hour)	-211	-150	Per hour	Government
150	Health	Practice nurse visit	-60	-40	Per visit	Government
151	Health	Home nurse visit	-141	-100	Per visit	Government

Impacts Database

Impact Inputs

Output Results

Output Results (Alt)

Output Summary

Assumptions

Wellbeing Impacts

Sensitivity Analysis

Previous Systems Week Example

Do Nothing

Do Nothing	Quantifying Option 14
Quantified Effect (cost): Reduced Wellbeing	
Assumption(s):	
A	The calculation is based on the entire New Zealand population.
B	A reduction of 0.05 points in life satisfaction is applied per person. (HM Treasury, 2021)
C	It is assumed that all individuals are negatively affected by plastic pollution.
D	Discount Rate is 8%
E	Time Frame is 10 years
F	Undiscounted Benefit is 0
Calculations:	
The do-nothing approach demonstrates that a 0.05-point reduction in wellbeing from plastic pollution, valued at \$5,212 per one-point change per person per year (CBAx tool (241)), results in an estimated annual loss of \$1.4 billion when applied to the New Zealand population (5,330,600 provisionally as of 2025). Over a 10-year period, this equates to a NPV of approximately \$12 billion [B].	
The cost for reduced wellbeing is approximately \$12 billion .	

Quantified Effect (cost): Damage to Marine Life	
Assumption(s):	
A	Global statistics used to calculate plastic waste that enters NZ oceans.
B	Discount Rate is 8%
C	Time Frame is 10 years
D	Undiscounted Benefit is 0
Calculations:	
Current estimates show New Zealand sends approximately 380,000 metric tonnes of plastic to landfills (Office of the Prime Minister's Chief Science Advisor & University of Auckland, 2019). Based on global estimates, approximately 0.5% of New Zealand's plastic waste ends up in the ocean (National Caucus of Environmental Legislators, n.d.). The cost of plastic waste per tonne has been quantified by Beaumont et al. (2019) as USD \$3,300 to \$33,000 per metric tonne of plastic waste. The annual cost was assumed to be USD \$15,000/tonne.	
Annual Cost Calculation:	
Approximate amount of plastic in oceans: 380,000 tonnes x 0.005 = 1900 tonnes	
Cost of damage to marine life: USD \$15,000/tonne x 1,900 tonnes = USD \$28.5 million	
USD to NZD (1.73NZD = 1.00USD)	
Cost of damage to marine life: NZD \$49.3 million	
The cost of damage to marine life is approximately \$50 million .	

Do Medium

Quantified Effect (benefit): Welfare gains from reduced littering	
Assumption(s):	
A	Litter Reduction = 8.3% (Davies, Barton, 2022)
B	Willingness-to-pay welfare gain figure of \$14 per household per year is based on a 14.5% litter reduction figure (Davies, Barton, 2022)
C	We can estimate a true willingness-to-pay by scaling it linearly based on the given litter reduction figures, i.e., 8.3% + 14.5%
D	There are 2,020,000 households in NZ (Stats NZ, 2024)
E	Welfare gains from reduced littering linearly increases up to the estimated benefit based on the number of machines we set up every year for the first 5 years.
Calculations:	
Based on the litter reduction figure from [A] and the scaled willingness-to-pay figure from [B] as per [C], we can calculate the corresponding maximum yearly benefit in these steps:	
- Scale factor = 8.3% + 14.5% = 0.57 - True welfare gain = Scale factor × \$14 per household per year = \$8.01 per household per year - Maximum yearly benefit = \$8.01 × 2.02 million households [D] = \$16.2 million per year	
The welfare gain from reduced littering is approximately \$16 million per year .	

Sometimes the best option is simply the least bad. (smallest negative NPV)

Limitations of the CBAX spreadsheet

- Not all impacts will be relevant to your projects.
- Not all impacts are listed in the database
 - Check the impacts database first, then move to other sources
- Organisations need to quantify impacts based on the best available evidence (sometimes it might not be the best evidence).
- Data sets can be incomplete (especially if trying something new).
- Gives exact dollar values, estimation unlikely to be that accurate (IMPORTANT FOR SYSTEMS WEEK).

Understand the assumptions behind the number not the number itself

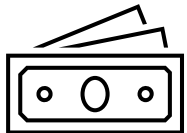
Step 5. Discount benefits and costs

Because these projects are very large and can span several years the **Time Value of Money** becomes very important.

You must **discount** future costs and benefits to today's value.



At systems level this is maximising the return for each dollar of taxpayers money spent



Calculating the discount rate

Project Level (ENGGEN 303)

Weighted average cost of capital (WACC)

$$WACC = \left(\frac{E}{V} \times Re \right) + \left(\frac{D}{V} \times Rd \times (1 - Tc) \right)$$

E = Market value of the firm's equity

D = Market value of the firm's debt

$V = E + D$

Re = Cost of equity

Rd = Cost of debt

Tc = Corporate tax rate

System Level (ENGGEN 403)

Social opportunity cost of capital (SOC)

Opportunity cost of private investment

Social rate of time preference (SRTTP)

Cost of benefits now vs in the future

$$\text{Social discount rate} = (\alpha)\text{SOC} + (1 - \alpha)\text{SRTTP}$$

No need to calculate use (8%)
see more on how these rates
are calculated [here](#)

Public Sector Discount Rates (Real)	Year 1-30	Year 31-100	Year 101+	Sensitivity test (mandatory)
Non-commercial proposals (SRTTP)	2%	1.5%	1%	8%
Commercial proposals (SOC)	8%	8%	8%	2%

Step 6. Assess the CBA

In practice is it worth pursuing the cost benefit analysis?

- Can be back of the envelope or very involved-research that can cost over \$1 million.
- Cost of conducting CBA must NOT exceed the potential value of the project. (CBA of the CBA)
- Who carries it out?
 - Policy advisors – Treasury, consultants
 - Decision makers (Government)
- Is more research required?



Step 7. Write the Report

How should these values be presented?

- Use summary tables: costs, benefits, and metrics. (refer to pg 41: *Guide to social cost benefit analysis*).
- In practice should have ranges or confidence intervals rather than point estimates. (point estimates fine for systems week)
- Appropriate accuracy. NOT to the nearest dollar! Usually 2SF!
 - Tens of thousands? Hundreds of thousands? Millions?
- If monetary costs outweigh monetary benefits but there are large intangible benefits, this MUST be highlighted to the reader in the summary.
- Should provide the intuition behind the result (tone, feeling).
- Sources and assumptions must be included in the appendix

Table 6. Summary of Cost-Benefit Analyses

	Do Nothing	Package 1	Package 2	Package 3
NPV Benefits	\$0	\$167 million	\$219 million	\$1 billion
NPV Costs	\$12 billion	\$67 million	\$108 million	\$491 million
NPV Net	-\$12 billion	\$100 million	\$110 million	\$542 million
BCR	N/A	2.5	2.0	2.1
IRR	N/A	59%	32%	70%

Limitations of the Cost Benefit Analysis

What happens if you change the timeframe of analysis?

- A 2-year horizon may produce a very different result from a 20-year horizon.
- What is the most appropriate timeframe, and why?

Short-term decision making can bias long-term outcomes.

- Government funding and election cycles often favour short-term thinking.
- Many of today's challenges are intergenerational and require a longer-term perspective.
- Indigenous worldviews often encourage thinking across multiple generations.

Costs and benefits can be difficult to measure and monetise.

- Particularly intangible, social, cultural, and environmental impacts (See Lecture 6).

Do financial values reflect societal values?

- CBA assigns monetary values to outcomes that different people value differently.
- Examples include equity, biodiversity, conservation, national infrastructure, and public versus private ownership.

Is the decision driven by values or politics, masquerading as a financial case?

- Assumptions, discount rates, and valuation methods can influence the preferred option.
- Political priorities may shape the analysis as much as the evidence.

Discounting the future, e.g. floods

A CBA for different flood mitigation options will be very sensitive to timeframe

- Flood banks?
- Raise buildings and reinforce bridges?
- Move the settlement to higher ground

The decision on timeframe of analysis may drive the decision.



Values, politics, choices

A CBA for a playground outside Parliament?

- Costs are easy to measure
- Benefits are all non-financial?
- How might you measure improvements to ... the culture of Parliament? attracting more talent as MPs? trust in institutions?
- What timeframe would you measure on?
- Attribution of benefits to the playground?



Values, politics, choices

CBA for the interisland ferries?

- What value would you place on a nationally integrated rail system?
- How do you weigh resilience to seismic events? climate change?
- What timeframe will you measure on? What will you measure? Against what counterfactual?





Key takeaways

- Identify, select, and rank quantifiable impacts by order of significance.
- Monetise these impacts using CBAX, comparable cases, studies, or other reports.
- Not all impacts will be monetizable or even quantifiable, but these qualitative impacts should still be communicated in the report.
- Compare non-monetary impacts using quantitative/qualitative analysis like MCA, or discussion.



Questions?

Thank you, and see you next
lecture 😊

